Interpretable Used-Tesla Asking Prices: Trim Indicators and Model-Specific Registration Effects

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https://github.com/v1852465cs15/nycu-data-mining-hw1-tesla

Abstract

Buyers and small dealers need comparable asking-price estimates when screening used electric vehicles. We study 3,754 European Tesla advertisements using a supplied 3,002/752 training/test partition. Ordinary least squares predicts price in euros from seven structured inputs, six title-derived trim indicators, and three model-by-registration-year interactions. The engineered model reduces dealer-grouped cross-validation MAE from EUR 3,991 to EUR 3,324, and supplied-test MAE from EUR 4,290 to EUR 3,711. A random forest remains better at EUR 3,227. Neither meets our proposed EUR 3,000 test-MAE threshold. Ablations isolate the value of each engineered group, and coefficient and case analyses expose sparse-category uncertainty and extreme-listing failures. Because the test set was inspected during earlier development, these are descriptive benchmark results, not a new blind evaluation.

1 Introduction

A Tesla model name alone does not identify its trim, software entitlement, or condition. A buyer screening advertisements needs a comparable-price reference before paying for inspection; a small dealer needs an initial asking-price check. We estimate the *advertised* price available in the observed market, not a sale price, future value, or certified fair value. An unusually large price gap should trigger inspection, not an automatic purchase decision.

Our question is whether inexpensive, listing-time trim indicators and model-specific registration-year slopes improve a transparent linear estimator beyond raw attributes and correlation ranking. We contribute (1) a fixed-split, train-only feature analysis (Sections 3–5), (2) matched linear ablations with dealer-grouped validation (Sections 6–7), and (3) final-model uncertainty and failure-case analysis (Section 8). The engineered model remains linear in its coefficients; we make no claim that it beats the nonlinear baseline.

2 Related Work

The studies below provide context; their markets, currencies and evaluation protocols differ, so scores are not directly comparable.

Study / data	Method / reported result
Pudaruth (2014) 97 Mauritian newspaper ads	10-fold kNN; MAE 45,189 rupees (Toyota), 25,741 (Nissan). Year and cylinder volume.
Bukvić et al. (2022) Croatian Njuškalo ads	Linear regression: MAE EUR 1,373.2; RMSE 2,104.7. Compared with RF/SVM; model, year and kilometres.
Madhusudhanan et al. (2024) About 2 million B2C records; 65 attributes	ProbSAINT: MAPE 5.6% / 5.3% on two temporal splits. Includes age, mileage and offer duration.
This study 3,754 European Tesla ads	Interpretable OLS with title flags and model-year interactions; test MAE EUR 3,711 on 752 rows.

Comparison: this study uses listing-time Tesla attributes and first registration, rather than manufacture or offer duration. All cited results retain their original units.

3 Task Definition and Data

One row is an AutoScout24 advertisement. The continuous target is its asking price in EUR. The training range is EUR 9,000–175,450; it defines observed coverage, not a clipping rule. Inputs are first-registration year, mileage (km), reported power (kW), equipment count, model, seller country, assistance label and selected title keywords.

All are available when screening a listing with its price hidden. Seller identity, URL, collection order and target-bearing free text are not predictors.

The supplied workbook records collection on 5 October 2026 across 188 result pages in eight European countries. It contains 3,754 listings, with the first 3,002 in training and 752 in testing. We retain this original partition for reproducibility. Search-page order is neither randomized sampling nor longitudinal forecasting; a later-market claim would require a temporal dataset. The collector's listing-ID deduplication does not establish unique physical vehicles.

Variable	Mean	SD	Range
Price (EUR)	35,308	13,937	9,000–175,450
Year	2021.46	2.43	2001–2026
Mileage (km)	88,835	58,372	9–635,000
Power (kW)	300	126	7–1,020
Equipment	55	22	2–119

Table 1: Training-only summary ($n = 3{,}002$). Inherited fields retain suspect values; range is observed, not a validity bound.

All seven structured inputs are populated in the supplied workbook, so no additional imputation is fitted. Earlier filling and assistance annotations are inherited, not independently verified. Numeric fields are standardized using the current training fold; categories are one-hot encoded with unknown categories ignored at prediction. We preserve price extremes, because unusually expensive cars need not be erroneous. One training Model Y has year 2001: the main benchmark retains it to match the shared data, and Section 7 reports an exclusion sensitivity. Reported power and equipment counts are seller fields, not certified specifications; equipment lists can include headings. Appearance and range are excluded from the primary model to avoid adding uncertain measurements without demonstrated incremental value.

4 Exploratory Feature Analysis

Figure 1 uses training rows only. Price is associated with mileage, year and vehicle segment, but each marginal plot mixes trims and countries. Training Pearson correlations with price are -0.403 for mileage, 0.286 for year and 0.557 for power. The year–mileage correlation is -0.594; this motivates examining redundancy rather than interpreting either slope in isolation. We retain power as a predictive seller annotation, while acknowledging its uncertain measurement basis.

Univariate numeric r^2 and categorical correlation ratio η^2 place power, model, mileage, year and country above equipment and assistance. These common-scale scores use raw EUR price, not log-price. We test three explicit hypotheses: H1, trim indicators reduce error conditional on structured fields; H2, model-specific year interactions improve on a shared slope; H3, these two groups outperform raw correlation ranking. Their verdicts come from matched ablations in Section 7 and coefficient interpretation in Section 8, not from these plots alone.

5 Method

5.1 Pipeline and feature groups

Figure 2 traces each listing from raw fields to a EUR prediction. Set A contains the seven structured fields. Set B retains the top k fields ranked inside each training fold: squared Pearson correlation for numeric fields and between-category/total target sum of squares for nominal fields. Searching $k = 1, \ldots, 7$, the smallest value within one standard error ($s/\sqrt{5}$) of the lowest CV MAE is $k = 5$, favoring a smaller representation when gains are within fold uncertainty.

Set C, our final OLS representation, keeps all seven raw fields and adds two groups. Six case-insensitive title flags mark Performance/Plaid, Long Range, AWD/Dual Motor, entry trim (RWD/Standard Range), new generation (Highland/Juniper/Plaid/refresh) and HW4. These capture within-model differences that power alone may miss (H1). Flags are mentions, not verified configurations: Plaid activates two flags, and Standard Range does not certify rear-wheel drive. The reproducibility package preserves exact expressions.

Three interactions allow Model S, X and Y to have different registration slopes:

$$u_m = \mathbf{1}(\text{model} = m)\,(\text{year} - \overline{\text{year}}_{\text{fit}}). \quad (1)$$

They address H2. The fit-partition mean prevents validation-dependent centering. Model 3 shares the base slope with rare models; these latter classes cannot support reliable separate slopes. This is a registration-year association, not observed longitudinal depreciation.

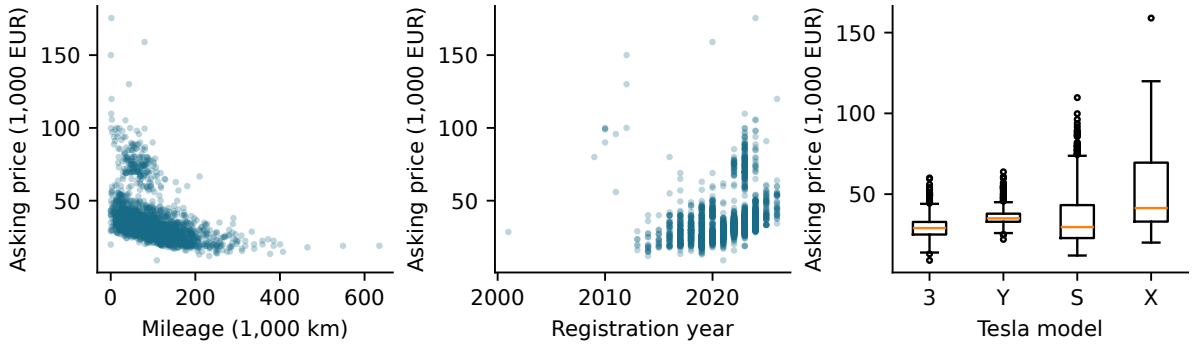


Figure 1: Training-only price against mileage, first-registration year and Tesla model (n = 3,002). Scatterplots retain extremes; boxes show quartiles and 1.5-IQR whiskers. Association is not causal depreciation.

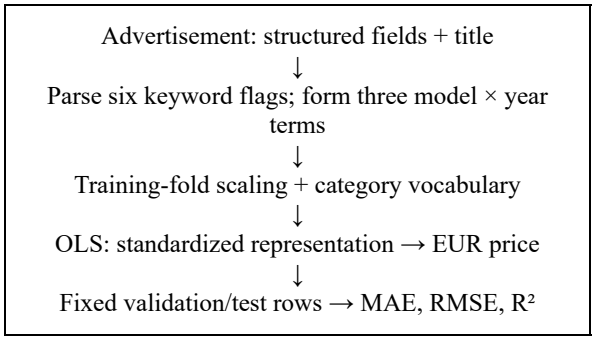


Figure 2: Prediction pipeline. Every learned transformation uses only its fit partition. The target enters regression fitting and evaluation, never feature construction.

5.2 Linear model and alternatives

OLS minimizes $\sum_i(y_i - b - z_i^T\beta)^2$. It offers direct EUR effects and a small fixed representation, while interactions allow non-additive raw-input behavior. It has no regularization parameter. No target logarithm, post-hoc clipping or higher-order polynomial search is used. Full one-hot fitting uses a least-squares solution; inference uses equivalent full-rank reference coding. The representation has 29 coefficients including the intercept.

Ridge is a plausible alternative for correlated features. A training-only dealer-grouped sensitivity searches $\alpha \in \{.01, .1, 1, 10, 100, 1000\}$ on Set C; the lowest MAE is EUR 3,324 at $\alpha = 0.01$, compared with OLS EUR 3,324. We retain the specified OLS benchmark without selecting a new model from test outcomes. Coefficient uncertainty is conditional on this representation.

6 Experimental Setup

6.1 Validation and provenance

We use identical five-fold shuffled KFold partitions (seed 42) across methods, plus five-fold GroupKFold by seller/dealer. Five folds retain about 80% for fitting while permitting a spread estimate. There are 1,749 training dealer groups; each missing seller receives its own group. Scaling, encoding, centering and top-k ranking are fitted within folds. We report mean ± sample SD. Feature-set/k exploration used these training folds, so CV is development evidence, not a nested estimate of the selection process.

The supplied test set was inspected in earlier team analyses. We freeze this integration's model definitions before recomputation and retain all 752 rows, but do not claim a new blind holdout. Dealer grouping is not a vehicle-identity guarantee; overlap and sensitivity checks are separate. A new locked test set is needed for confirmatory claims.

6.2 Metrics and baselines

MAE measures absolute EUR error; RMSE penalizes large misses, and R^2 describes explained variation. EUR 3,000 is an **assumed benchmark tolerance**, not an inspection fee or validated buyer/dealer cost. No stakeholder cost data were collected. Pass/fail and within-threshold shares are descriptive; they do not establish economic benefit or an individual guarantee.

Baselines are the **Trivial Baseline (training mean)**, **Simple Baseline (mileage-only OLS)**, and **Strong Baseline (200-tree random forest on Set A, seed 42; no depth limit, default leaf size 1)**.

The forest is an inherited fixed flexible comparator, not a tuned optimum. All methods use identical rows and metrics. Paired test-loss gaps use 10,000 dealer-cluster bootstrap resamples (seed 42), keeping both methods' predictions together. Grouped-fold paired t-tests have only five overlapping training fits, so their p-values are exploratory.

7 Results

7.1 Main comparison

Method	Group MAE	Test MAE	RMSE	R^2
Mean	8,552 ± 1,354	8,906	14,040	−0.004
Mileage	7,806 ± 1,171	8,116	12,909	0.152
RF (raw)	**2,660** ± 236	**3,227**	**5,103**	**0.867**
OLS A	3,991 ± 429	4,290	5,971	0.818
OLS C (ours)	3,324 ± 267	3,711	5,399	0.852

Table 2: Main results: errors in EUR; Group MAE is five-fold mean ± sample SD. RMSE and R^2 use all 752 test rows. Bold marks the best values.

Table 2 shows that engineered OLS improves over raw OLS but remains behind the random forest. Test MAE falls by EUR 578 (13.5%) from A to C; the forest's advantage over C is EUR 484. A piecewise tree ensemble can capture local trim/usage combinations beyond the three slopes and six flags, although the comparison does not isolate that mechanism. The forest remains a baseline; our final predictor is linear.

Against the proposed threshold, C exceeds EUR 3,000 by EUR 711, and RF exceeds it by EUR 227. Only 55.6% of C's test errors lie within EUR 3,000 (RF: 59.4%). The average threshold is not satisfied even though many individual estimates fall inside it. Thus these predictions are inspection aids, not autonomous pricing decisions.

7.2 Ablations and uncertainty

Only the feature representation changes in Table 3; every row uses OLS and the same splits. Removing the model×year interactions raises KFold MAE by EUR 462 ($p = 0.0001$) and grouped MAE by EUR 455 ($p = 0.004$); removing the flags raises them by EUR 213 ($p = 0.0003$) and EUR 198 ($p = 0.002$), so each group helps when the other is present. Set C also improves on top-k Set B: ranking raw associations cannot create title information or model-specific slopes. Both groups reuse collected fields; their costs are parsing maintenance and harder interpretation of overlapping trim mentions.

Set	KFold MAE	Group MAE	Test MAE
A: raw	3,939 ± 105	3,991 ± 429	4,290
B: top-5	3,966 ± 118	4,022 ± 446	4,329
C: full	**3,244** ± 92	**3,324** ± 267	**3,711**
C − flags	3,457 ± 95	3,522 ± 278	3,871
Δ (p)	*+213 (.0003)*	*+198 (.002)*	*+160*
C − interactions	3,705 ± 58	3,779 ± 414	4,081
Δ (p)	*+462 (.0001)*	*+455 (.004)*	*+370*

Table 3: Matched OLS ablations, EUR. All folds and preprocessing rules are shared. Lower is better; C is best in every column. Δ: MAE increase when the group is removed from C; p: exploratory paired fold t-test.

Comparison	Gap	95% CI	p
A − C	578	[417, 742]	0.001
B − C	617	[453, 783]	0.002
RF − C	−484	[−723, −236]	0.001

Table 4: Test-MAE gaps (EUR): positive favors C. CI: paired bootstrap over the 752 test cars, 10,000 draws; p: exploratory paired grouped-fold t-test.

Table 4 uses Part C's car-level paired bootstrap, distinct from the dealer-cluster procedure in Section 6.2. Both keep predictions paired. These fixed-fit intervals omit retraining, selection and market change; reused test data limits prospective claims.

Model	n	A	B	C	Gain	Med. %	≤3k
3	347	3,467	3,481	3,004	−13%	7.9%	63%
Y	271	3,723	3,770	3,161	−15%	6.7%	58%
S	69	6,790	6,917	6,289	−7%	16.7%	29%
X	59	7,816	7,876	6,423	−18%	9.5%	42%
Roadster	5	12,403	12,395	11,976	−3%	13.0%	0%
Cybertruck	1	22,051	21,858	19,203	−13%	12.0%	0%
All	752	4,290	4,329	3,711	−13%	8.0%	56%

Table 5: Test MAE by model (EUR). Gain: C vs A; Med. %: C's median absolute % error; ≤3k: C's share within EUR 3,000. Grey: too few rows.

C helps every common model (Table 5), least for S. Relative to price, X (9.5%) is close to Model 3 and Y, while S is weakest (16.7%) because most Model S listings are old, cheap cars. Rare classes are not interpreted.

No model meets the EUR 3,000 threshold: 47% (A), 56% (C) and 59% (RF) of test cars lie within it. MAE exceeds it by EUR 1,290, 711 and 227. Excluding six rare cars gives EUR 4,211, 3,635 and 3,090. Model 3's EUR 3,004 also exceeds the threshold, by EUR 4.

Sensitivity to data quality. A duplicate proxy (model, registration month, mileage) flags 152 test rows overlapping training; on the other 600, MAE

is EUR 4,550 (A), EUR 4,031 (C) and EUR 3,816 (RF). Matching attributes do not prove identical vehicles. Dropping the Model Y/2001 training record gives C test MAE EUR 3,713. Neither post-hoc audit changes the headline model.

8 Analysis

8.1 Coefficients and significance

Term	β	95% CI	p	Rows	Natural unit
Year	4.56	[4.06, 5.07]	<.001	3,002	+1,882/yr
Mileage	−3.44	[−3.87, −3.02]	<.001	3,002	−590/10k km
Power	2.78	[2.31, 3.24]	<.001	3,002	+220/10 kW
Equipment	0.19	[−0.03, 0.41]	.09	3,002	+9/item
Performance	2.94	[2.17, 3.71]	<.001	619	+2,941
Long Range	1.09	[0.39, 1.79]	.002	1,143	+1,087
AWD	−1.46	[−2.23, −0.69]	<.001	1,497	−1,461
Entry trim	−1.39	[−2.08, −0.70]	<.001	780	−1,387
New gen.	4.35	[3.24, 5.46]	<.001	177	+4,348
HW4	5.20	[3.35, 7.06]	<.001	62	+5,204
Model S×year	3.17	[2.62, 3.72]	<.001	380	+2,296/yr
Model X×year	2.96	[2.36, 3.57]	<.001	333	+3,112/yr
Model Y×year	−0.27	[−1.13, 0.59]	.54	1,091	−277/yr
Model S	20.31	[18.83, 21.79]	<.001	380	+20,308
Model X	24.76	[22.84, 26.68]	<.001	333	+24,759
Model Y	3.24	[1.90, 4.59]	<.001	1,091	+3,242
Roadster	85.02	[67.31, 102.74]	<.001	11	+85,023
Cybertruck	133.40	–	–	1	+133,400
Unknown	−11.69	[−20.40, −2.98]	.008	2	−11,691
Italy	2.78	[2.09, 3.46]	<.001	373	+2,775
Spain	−1.43	[−2.22, −0.64]	<.001	326	−1,430
Belgium	−0.84	[−1.56, −0.12]	.02	294	−842
Germany	−0.39	[−1.01, 0.23]	.21	685	−393
France	−0.29	[−1.15, 0.58]	.51	155	−288
Austria	−0.09	[−0.76, 0.58]	.79	268	−93
Luxembourg	−0.10	[−2.76, 2.56]	.94	7	−99
FSD	0.87	[0.14, 1.60]	.02	293	+866
EAP	0.60	[−0.32, 1.52]	.20	135	+599

Table 6: Final Set C coefficients, HC1 95% CI; β, CI in 1,000 EUR (continuous: per SD; flags: vs 0; categories: vs Model 3, Netherlands, AP). Natural unit in EUR. Grey: <20 training rows, not interpreted; the one Cybertruck is fitted exactly.

Table 6 lists all 28 non-intercept terms. HC3 is undefined (one leverage-1 row, the single Cybertruck), so we use HC1 robust t inference. Training R^2 is 0.850; max VIF 5.9 (year), moderate. HC1 ignores dealer dependence and no multiple-testing correction is applied, so results are exploratory associations.

Interpretation. Each newer year adds EUR 1,882 for Model 3, about EUR 4,178 for S and EUR 4,994 for X; Y does not differ from Model 3. This supports heterogeneous slopes, not a causal effect of aging. Trim matters beyond power and model: HW4, new generation, Performance and Long Range carry premiums and the entry trim a discount (Table 6). Italy (+EUR 2,775), Spain (−EUR 1,430) and Belgium (−EUR 842) differ from the Netherlands; other countries do not. FSD adds EUR 866; EAP is not significant. AWD (−EUR 1,461) corrects for overlap: 88% of AWD-flagged listings also carry Long Range or Performance and report more power (318 vs 282 kW), so those terms take most of the premium; a Performance AWD car still nets +EUR 1,480. Raw AWD listings are also cheaper (EUR 34,596 vs EUR 36,016), so the sign is not a reversal.

	Hypothesis	Test	Δ	p	Verdict
H1	Newer, lower km, more kW cost more; models differ	HC1 t	–	<.001	Supported
H2	Colour adds nothing	+colour	+32	.32	No gain
H2	Country adds nothing	−country	+78	.010	Rejected, small
H3	Car model adds little	−model	+2,149	<.001	Rejected
H4	Power effect varies with kW	+power^2	−107	<.001	Supported

Table 7: Hypothesis verdicts. Δ: KFold MAE change vs C (EUR 3,244), positive = worse; p: paired fold t-test (H1: coefficient t-tests).

Resolving the hypotheses. H1 concerns effect directions, so it is read from the coefficients: all point the expected way with $p < .001$. H2–H4 add or remove one group and compare KFold MAE with C (paired fold t-test). Colour gives no prediction gain, although a joint Wald test finds a small in-sample link. Country matters, but little (+EUR 78, $p = 0.010$). Removing car model, with its year terms, costs EUR 2,149, by far the largest change. Price is curved in power (−EUR 107, $p < .001$); we report this without changing the frozen model. A small p-value does not make an effect useful: compare its natural-unit size to the EUR 3,000 screening tolerance.

8.2 Error analysis

#	Listing	Actual	Pred.	Error	Likely cause
1	X 75D, FR, 2018	75,900	34,842	41,058	Mispricing
2	X 75 DM, BE, 2023	90,000	61,820	28,180	Inconsistent inputs
3	S 85, NL, 2014	12,000	−15,506	27,506	Extrapolation
4	S 70, ES, 2016	48,600	24,643	23,957	Mispricing
5	S P85 RWD, ES, 2013	32,990	9,683	23,307	Flag failure + extrap.

Table 8: Five largest absolute errors of final Set C (EUR; error = actual − predicted).

Cases 1 and 4 are priced far above comparable training listings (27 Model X "75": max. EUR 39,999; nine Model S 70: max. EUR 34,000), so the listings, not the predictions, look off. In Case 2 the title and the listing's own 244 kW describe an old 75D, but registration says 2023. Case

3's 670,000 km exceeds the training maximum (635,000 km): linear mileage and the steep S slope give C's only negative prediction. Case 5 has a normal price (comparable median EUR 26,170), but the Performance pattern misses "P85", "RWD" turns on the entry-trim flag, and the S slope pushes the 2013 car down.

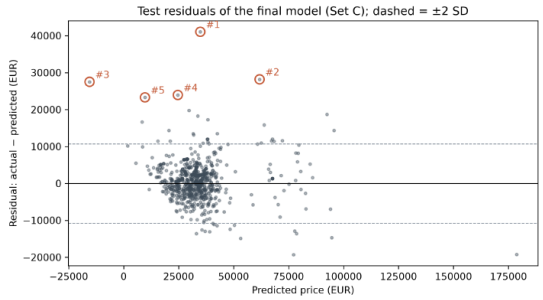


Figure 3: Set C test residuals, actual minus predicted EUR; dashed lines at ±2 SD. Circles mark the five cases in Table 8; all 752 test rows are shown.

Shared pattern. Old Model S cars are the main weakness (Cases 3–5; 16.7% median error): the model×year term assumes a constant euro loss per year, too steep for the oldest cars. 70% of the 20 largest errors are under-predictions (residual skew 1.14; Figure 3). First fixes: a flattening age term per model, a minimum-price clip, a "P85"/"P90" Performance pattern, and checks for disagreeing title, power and year. They came from test errors, so they are not applied, to avoid tuning on the test set.

9 Conclusion and Limitations

Title indicators and model-specific registration slopes improve this linear benchmark: test MAE is EUR 3,711, versus EUR 4,290 for raw OLS and EUR 3,227 for RF. Ablations support both engineered groups and improvement over raw correlation ranking. The gain is useful but insufficient for the proposed EUR 3,000 average-error threshold. Buyers should use the estimate to prioritize inspections and verify trim, battery condition and anomalous specifications before negotiating.

Coverage is one website, eight countries and one asking-price snapshot. The supplied order split, test exposure, possible duplicates, inherited annotations, rare classes and seller measurements limit generalization. EUR 3,000 is an assumed tolerance, not an established cost; it is relatively stricter for cheaper cars. Linear extrapolation can be implausible; associations are not causal depreciation. The next experiment should freeze these rules, collect a new locked snapshot with vehicle-identity and battery-condition checks, and evaluate both price accuracy and actual screening decisions.

Contributions of Each Member

Group A — 馮昱祥 (515559007), 林暉庭 (314551160): jointly prepared the abstract and introduction, defined the problem and research question, reviewed related work, collected and preprocessed the data, conducted exploratory feature analysis, and designed the feature-engineering pipeline and its figure.

Group B — 杜洛鋒 (315554001), 邱育宏 (315554053), 吳宥駒 (315551146): jointly handled model choice and hyperparameter tuning, validation protocol, metrics, decision threshold and baselines, and the main model comparisons with significance testing.

Group C — 黃少鋒 (315561001), 莊誌強 (315561002): jointly ran feature ablation, interpreted coefficients with statistical significance and VIF, analysed the largest errors, and wrote the conclusion and limitations.

AI usage. ChatGPT/Codex assisted with code revision, experimental recomputation, plots, literature checking, writing and compilation. Team packages supplied the data and feature definitions; the team retains responsibility for verifying the integrated results.

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